

Wireshark-traffic-investigation

Purpose

The aim of this study is to examine the network traffic captured by using Wireshark and to find out active protocols, websites visited, communication patterns, and encrypted traffic. This project aims to mimic the tasks of an SOC (Security Operations Center) analyst by capturing real-time network traffic data.

Methodology

Network traffic data is collected using Wireshark within 3-5 minutes while doing various activities on the internet including visiting Google, YouTube, GitHub, and pinging google.com website. Later, filters and analysis techniques of Wireshark like DNS filter, protocol tree, conversations, and ICMP analysis are used to analyze the traffic data patterns.

Tools Used

- Wireshark
- Web Browser (Chrome/Edge)
- Command Prompt (ping command)

Analysis and Results

A. Websites Accessed (DNS Analysis)

DNS queries were found to provide access to the websites listed below: google.com, youtube.com, github.com in addition to some background queries for services from Google.

B. Network Protocols Used

Analysis of network traffic indicated that the following protocols were used: TCP, UDP, DNS, TLS, and ICMP.

C. Network Conversations

Most of the communication took place between the host computer and the external servers used by Google and other cloud services mainly on HTTPS connections.

D. Encrypted Communication

Encrypted communication was evident in the form of TLS-encrypted data packets. Handshake packets including Client Hello and Server Hello were present in the packet capture.

E. ICMP Communication

Packets of ICMP protocol related to Echo Request and Echo Reply were found to be in presence in the captured packets confirming successful operation of the ping google.com command.

F. Analysis of External Ips

One of the external IPs analyzed in the Conversation tab turned out to belong to a genuine cloud service provider (Google/Cloudflare/Microsoft). suggesting legitimate background network communication.

Conclusion

Through this experiment, it was possible to show how it is possible to capture and analyze the network traffic by using Wireshark. DNS queries, protocols, encrypted traffic, and ICMP packets were successfully found. It has been proved that modern Internet traffic is mainly encrypted with TLS protocol and consists of several protocols.